

Discriminant Analysis

Backhaus K, B Erichon, S. Gensler, R. Weiber, T. Weiber, Multivariate Analysis – An Application Oriented Introduction, 2nd ed., Springer Gabler, 2023

Johnson, R. A., Wishern, D. W., Applied Multivariate Statistical Analysis, sixth ed., Pearson, 2024

Introduction:

- ❖ Whether multiple **causes** that are **metric** in nature influence **two or more different mutually exclusive outcomes**
- ❖ How do voters choose different political parties?
- ❖ Draw a random sample of voters of the different parties, and collect socio-demographic, **psychographic**, and attitudinal data.
- ❖ **Psychographic**: Study of people's **psychological** traits and behaviors - their **attitudes, values, and interests**. It's a market research method that helps businesses understand their customers and target them more effectively

Variables

- ❖ The party of choice by the voter is a **categorical (nominal)** variable.
- ❖ The party choice nominal variable may take values of different categories that are mutually exclusive (**ME**).
- ❖ **ME** means that is, each person can be assigned to choose one specific party.
- ❖ The **metric** variables used to correlate with the decision i.e., influence the decision may be voters age, income, **consumption orientation**, or attitude towards technology.

Process of Discrimination

- ❖ Variables: Attitude towards technology, age, income, and consumption orientation
- ❖ These variables are assumed to be **metrically scaled**
- ❖ With the help of **discriminant analysis**, we identify **which of the variables** that describe the voters **discriminate** the supporters of different parties
- ❖ **Discrimination:** Process of making distinctions between people based on the groups, classes, or other categories to which they belong or are perceived to belong such as race, gender, age, class, or religion
- ❖ Data analyst chooses which of the describing variables are relevant/significant

Process of Discrimination

- ❖ Discriminant analysis: Multivariate data analysis technique
- ❖ Analyze the relationship between **ONE categorical dependent variable** and a set of **metric (normally distributed) independent variables**
- ❖ The categorical variable (CV) is the grouping variable as its values leads to differentiating between the various groups
- ❖ CV reflects the group (e.g., party) an observation (i.e., voter) belongs to,
- ❖ Other examples, are buyers of different brands, or firms with different performances.
- ❖ For just two categoric groups, the technique is called **two-group discriminant analysis**
- ❖ For > 2 categoric groups, the technique is **multi-group discriminant analysis**

Application of Discriminant Analysis

- ❖ We can apply discriminant analysis to identify describing (independent) variables that discriminate between different groups
- ❖ It is also used to assess the discriminatory power of the describing variables (discrimination task).
- ❖ Next, we apply discriminant analysis to predict the group membership of new observations based on the describing variables once we know which describing variables distinguish the members of the groups (classification task).

Example: Credit Scoring

- ❖ Customers of a bank who have taken a loan may be divided into 'good' and 'bad' customers according to their payment behavior.
- ❖ Using DA, we examine which variables (e.g., age, marital status, income, duration of current employment, or number of existing loans) differ between the two groups
- ❖ Thus, we identify the **set of discriminatory variables**
- ❖ When a new customer applies for a loan, the bank predicts his/her creditworthiness based on his/her attributes viz. age, marital status, income, duration of current employment, or number of existing loans

Six Step Procedure in DA: First Step

- ❖ We first **define the groups** and, thus, **the categories of the dependent variable**
- ❖ We then specify **one or more discriminant functions**, which are **linear combinations of the independent variables** that best discriminate between the groups.
- ❖ If the grouping variable represents just 2 groups, we specify 1 discriminant function.
- ❖ If we observe three or more groups, we use **more than one** discriminant function.

Six Step Procedure: Further Steps

2. We estimate the discriminant function(s).
3. We assess the discriminatory power of the estimated discriminant function(s)
3. We examine the discriminatory power of the describing (independent) variables and assess which describing variables contribute the most to explaining the differences between the groups.
3. We describe how (new) observations are classified to one of the groups based on the observed values of the describing variables.
3. Finally, we test the assumptions of the discriminant analysis.

Six Step Procedure: Further Steps

- ❖ A manager of a chocolate company wants to know whether the buyers of its own brand (here: focal brand) perceive its chocolates differently compared to the buyers of its main competitor brand.
- ❖ The manager considers two describing variables to be relevant: 'price' and 'delicious'.
- ❖ We examine 12 buyers of the focal brand and 12 buyers of the main competitor brand.
- ❖ All 24 respondents provided information about their perceptions using a 7-point scale (from 1 = 'low' to 7 = 'high').

Perceptions of buyers of the focal brand and the main competitor brand (example case)

Buyers of focal brand (group = 1)			Buyers of main competitor (group = 2)		
Buyer	Price	Delicious	Buyer	Price	Delicious
1	2	3	13	5	4
2	3	4	14	4	3
3	6	5	15	7	5
4	4	4	16	3	3
5	3	2	17	4	4
6	4	7	18	5	2
7	3	5	19	4	2
8	2	4	20	5	5
9	5	6	21	6	7
10	3	6	22	5	3
11	3	3	23	6	4
12	4	5	24	6	6

Definition of Groups

- ❖ The grouping variable must be a categorical variable
- ❖ It should reflect different mutually exclusive and collectively exhaustive groups.
- ❖ We can also convert a metric variable (e.g., a firm's profit) into a categorical variable (e.g., low vs. high performance) to form the grouping variable.
- ❖ However, we need to be aware that we lose information if we do so.

Definition of Groups

- ❖ Generally, the **number of groups should not be larger than the number of describing variables**.
- ❖ In the example above, we have two describing variables, namely 'price' and
- ❖ 'delicious'.
- ❖ Thus, we restrict the example to two groups: 'buyers of the focal brand' and 'buyers of the competing brand'.
- ❖ The groups are identified by a group index g ($g = 1, 2, \dots, G$), where G is the total number of groups (here: $G = 2$ and $g = 1, 2$).

Recommendations related to the minimum number of observations

- ❖ It is important to note that the **reliability** of the discriminant analysis depends on the **number of observations**.
- ❖ We should have at least five observations per describing variable; 20 observations per describing variable are recommended.
- ❖ Moreover, each group should have at least 20 observations to warrant statistically significant and reliable results.
- ❖ Finally, the relative sizes of the groups should be comparable.

Number of Observations Within the Group

- ❖ If for all features **within a group**, the number of entries is the same, say 10, the number of observations n is equal to number of entries, i.e., 10
- ❖ But, if the number of entries for the features within a group is different number of entries, say 10 and 12, what is then the total number of observations?
- ❖ The common answer is that the features may not have missing information.
Either delete extra data or impute missing data seems to be the consensus.

We will again look more closely when we handle actual data to see why this is true

Number of Observations Across the Groups

- ❖ If for all features **across groups**, the number of entries may be different, say 10 in group 1 and 12 in group 2
- ❖ However, the number of observations may not be too different across groups
- ❖ The condition is all features should be represented in each of the groups
- ❖ All groups should have the same features

Recommendations related to the minimum number of observations

- ❖ Large differences in the relative group sizes will influence the estimation of the discriminant function(s) and the classification of the observations.
- ❖ Regarding these considerations, the groups in our example are actually rather small (only 12 observations for each group), but they are just meant as an illustrative example.

Participants Doubt: Ordinal to Metric

In choosing a chocolate brand, customers rate taste and price in a 1 to 7 scale which is an ordinal variable. How can it be used as a metric variable in linear discriminant analysis?

- a) Treat 1 - 7 ordinal scale as a metric variable by directly using the numeric values (1 through 7) assuming the steps between numbers represent equal intervals.
- b) We assume 7 is more and 1 is less (rating not ranking)
- c) Scales with 5 or more categories (like 1 to 7) behave very similarly to continuous interval data in statistical practice
- d) Option available to mathematically convert ordinal categories into optimized metric values before applying LDA (needs to be explored)

Participants Doubt: Ordinal to Metric

In choosing a chocolate brand, customers rate taste and price in a 1 to 7 scale which is an ordinal variable. How can it be used as a metric variable in linear discriminant analysis?

- a) It is obviously an approximation
- b) There are several ordered categories (e.g., 7 rather than only 2–3).
- c) The categories are reasonably evenly spaced.
- d) The distribution is not extremely skewed. If the rating distribution for both taste and price is unimodal, relatively symmetrical, and lacks extreme skewness
- e) The sample size is reasonably large.
- f) The analysis is intended to capture an approximately linear relationship.

Participants Doubt: Clustering vs. LDA

The main difference is that clustering is an **unsupervised** learning method that groups data without any pre-defined labels, while discriminant analysis is a **supervised** learning method that builds a model to classify items into pre-existing, known groups

Discriminant analysis is supervised (requires a training dataset with known group memberships).

Group Definition: In clustering, the number of groups and their identities are unknown ahead of time.

In discriminant analysis, the groups are completely defined before the analysis begins.

Primary Goal: Clustering aims to discover natural, hidden structures or segments in data.

Discriminant analysis aims to find prediction equations or rules that best separate known groups and classify new observations.

Participants Doubt: Clustering vs. LDA

Cluster Analysis: It measures the distance or similarity between data points to maximize internal similarity (within groups) and external difference (between groups).

Common use case: Customer segmentation or finding natural groupings in unlabelled survey data.

Discriminant Analysis: Uses a known dependent grouping variable to find which independent variables best distinguish between the categories.

How it works: It analyzes a training set where group membership is already known, creating a mathematical function to separate the classes.

Common use case: Predicting whether a student will succeed based on past records or diagnosing medical outcomes from historical patient data.

The Discriminant Function

The discriminant function is a linear combination of the describing variables:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_j X_j$$

Where

Y: dependent variable (grouping variable)

- b_0 : constant term
- b_j : discriminant coefficient of the independent (describing) variable j
- X_j : independent (describing) variable j

Discriminant function

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_j X_j$$

- ❖ The discriminant function is also called **canonical discriminant function**
- ❖ The discriminant variable Y is called canonical variable
- ❖ The term canonical indicates that the variables are combined linearly.
- ❖ It is thus assumed that the relationship between the independent and dependent variables is linear.
- ❖ For each observation that is described by some variables, the discriminant function predicts a value for the discriminant variable Y .

Discriminant function

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_j X_j$$

- ❖ The discriminant function is also called **canonical discriminant function**
- ❖ In this context, "canonical" refers to the fact that the discriminant function is derived in **a standardized, optimal mathematical form**
- ❖ The discriminant variable Y is called canonical variable
- ❖ The term canonical indicates that the variables are combined linearly

Discriminant function

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_j X_j$$

- ❖ The discriminant variable Y is called canonical variable
- ❖ In canonical discriminant analysis, canonical means that the analysis finds special (optimal) linear combinations of the original variables that maximize the separation among the groups and are mutually uncorrelated (orthogonal)

Thus, if there are several canonical discriminant functions, they are of the form

$$D_1 = a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p,$$

$$D_2 = a_{21}X_1 + a_{22}X_2 + \dots + a_{2p}X_p,$$

Discriminant function

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$$D_2 = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p,$$

where each function is a different optimal linear combination of the original features.

So, the phrase "**the variables are combined linearly**" means that the canonical discriminant function is formed by **adding weighted versions of the original variables**, not by multiplying variables together or using nonlinear transformations. The term **canonical** emphasizes that these linear combinations are chosen to be the best ones for discriminating among the groups.

Canonical Linear Discriminant function

- ❖ The term signals that this particular linear combination isn't arbitrary —
- ❖ it's the optimal one, derived mathematically
- ❖ Among all possible linear combinations of the variables, the canonical discriminant function is the one that best separates the groups.
- ❖ If there are more than two groups, additional canonical functions (orthogonal to the first) can be extracted, each capturing progressively less of the remaining group separation — analogous to how canonical correlation or principal components extract successive "best" linear combinations.
- ❖ In short: "Linear" tells you the mathematical form (a weighted sum, no curves or interactions), and "canonical" tells you that this specific weighted sum is the **statistically optimal one** for discriminating between the groups.

Why Linear Matters?

The function only involves each variable raised to the first power, multiplied by a constant weight and then **summed**. It does *not* include:

- Squared or higher-power terms (X_1^2 , X_2^3 , etc.)
- Interaction/product terms ($X_1 \times X_2$)
- Any other nonlinear transformation (logs, ratios, exponents, etc.)

So geometrically, the discriminant function defines a straight line (in 2D), a flat plane (in 3D), or a flat hyperplane (in higher dimensions) that separates the groups — rather than a curved boundary.

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Why Linear Matters?

Linear Regression

A linear regression model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon.$$

The model is called **linear** because it is **linear in the unknown coefficients** $\beta_0, \beta_1, \dots, \beta_p$.

Notice that each coefficient appears only to the first power and is not multiplied by another coefficient.

The predictors themselves **need not appear linearly**. For example,

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

is still a **linear regression model**, even though X^2 is a nonlinear transformation of the predictor. The model remains linear because it is linear in the coefficients β_1 and β_2 .

❖ In Linear Regression we focus on the linearity of the regression coefficients

❖ In Linear Discriminant Analysis, we focus on the linear combination of the predictors/independent variables

Why Linear Matters?

- ❖ In Linear Regression we focus on the linearity of the regression coefficients
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Discriminant function

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_J X_J$$

- ❖ The coefficients b_0 and b_j ($j = 1, 2, \dots, J$) are estimated based on the observed data in such a way that the **groups differ as much as possible with respect to the values of the discriminant variable Y**.
- ❖ Since the describing variables are metric, the resulting discriminant variable Y is also metric and **does not directly indicate the group membership**
- ❖ How estimates of group membership are derived from the estimated value of the discriminant variables such as Y forms the scope for succeeding discussion

Estimation of the Discriminant Function

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_j X_j$$

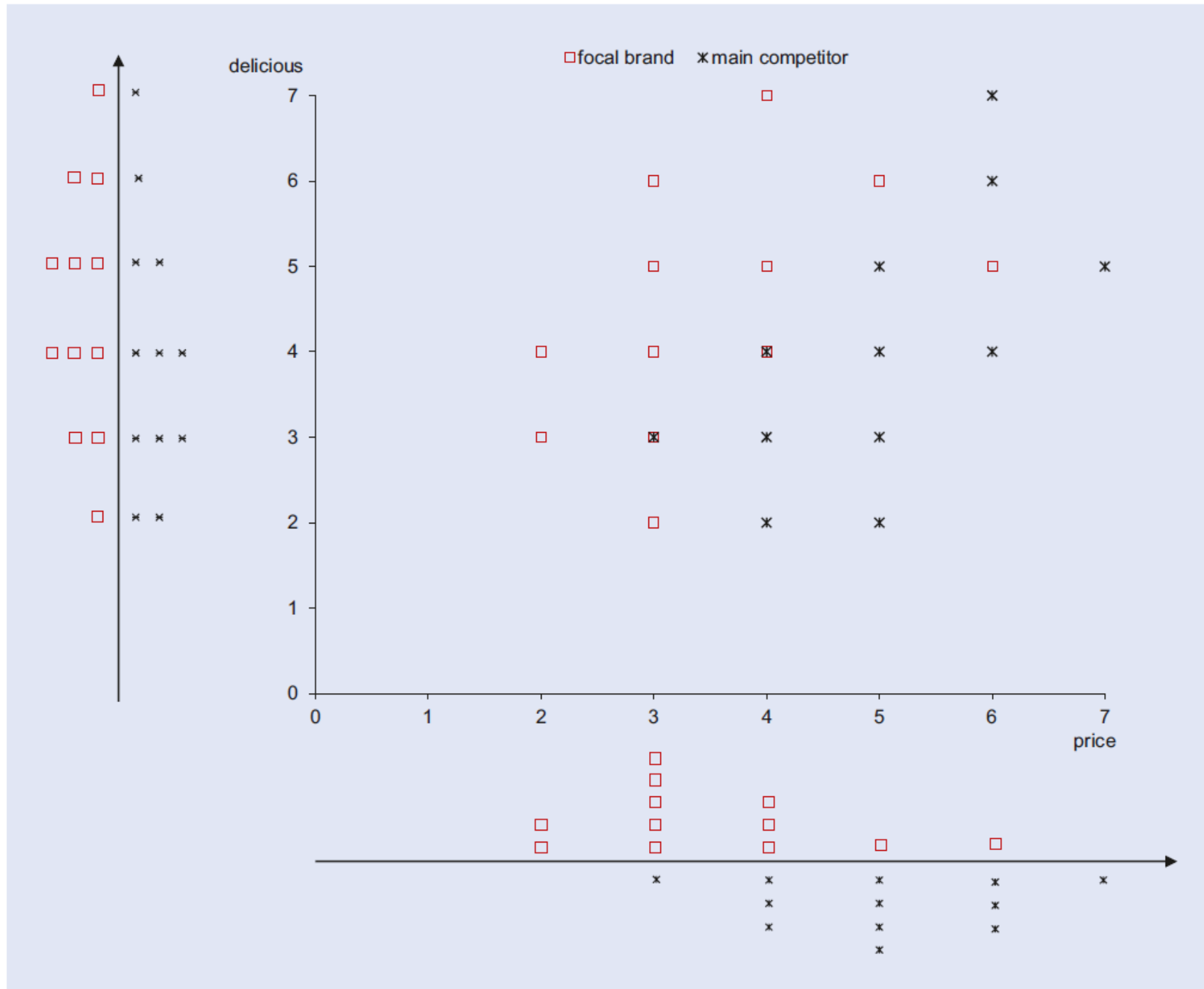
- ❖ When estimating the unknown coefficients b_0 and b_j of the discriminant function, the objective is to identify all b_j in such a way that they separate the groups well.

This requires an

1. **objective function / criterion**
2. that **maximizes** the
3. **separation of the groups (discriminant criterion).**

Discriminant Criterion

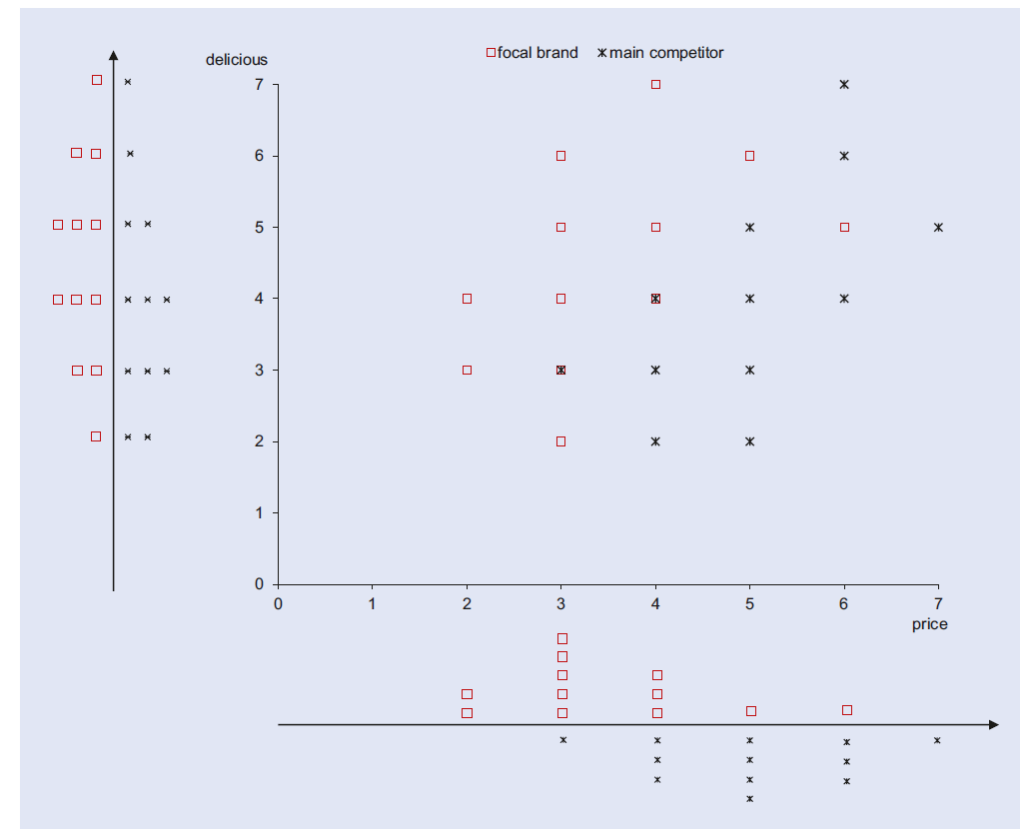
- ❖ Discriminant analysis aims to identify the describing variables that discriminate the groups.
- ❖ This implies that the groups differ with respect to the describing variables
- ❖ If they do so, we should observe different values for the describing variables in the two groups of the example presented above.
- ❖ Figure below displays a scatterplot of the observed values of the describing variables.
- ❖ The buyers of the focal brand are represented by red squares and the buyers of the



Scatter Plot
of the
Observed
Data

Discriminant Criterion

- ❖ Figure also shows the frequency distributions (histograms) of the values of the describing variables (i.e. 'price' and 'delicious') below ('price') or beside ('delicious') the scatterplot.
- ❖ The frequency distribution for each group is displayed separately, and the axes correspond to the original x- and y-axes.
- ❖ We learn that buyers of the competing brand tend to rate 'price' higher than buyers of the focal brand (cf. histogram below the scatterplot).



Discriminant Criterion

- ❖ The respective mean values of 'price' are 5.0 (competitor) and 3.5 (focal).
- ❖ In contrast, buyers of the focal brand rate, on average, rate 'delicious' higher than the buyers of the main competitor brand (mean values: 4.5 focal vs. 4.0 competitor) –refer histogram beside the scatterplot
- ❖ However, due to the significant overlaps of the two distributions, neither variable seems to separate the two groups very well
- ❖ Visual inspection of the histograms suggests that 'price' may separate the groups better because the groups seem to differ more with respect to this variable

Mathematical Theory: Basics

(1) separating two classes of objects or

(2) assigning a new object to one of two classes.

❖ It is convenient to label the classes π_1 and π_2 and the categories are ordinarily separated or classified on the basis of measurements on, for instance, p associated random variables $\mathbf{X}' = [X_1, X_2, \dots, X_p]$

❖ The observed values of $\mathbf{X}$ differ to some extent from one class to the other.

❖ We can think of the totality of values from the first class as being the population of $\mathbf{x}$ values for π_1 and those from the second class as the population of $\mathbf{x}$ values for π_2

Alternate Approach: Basics

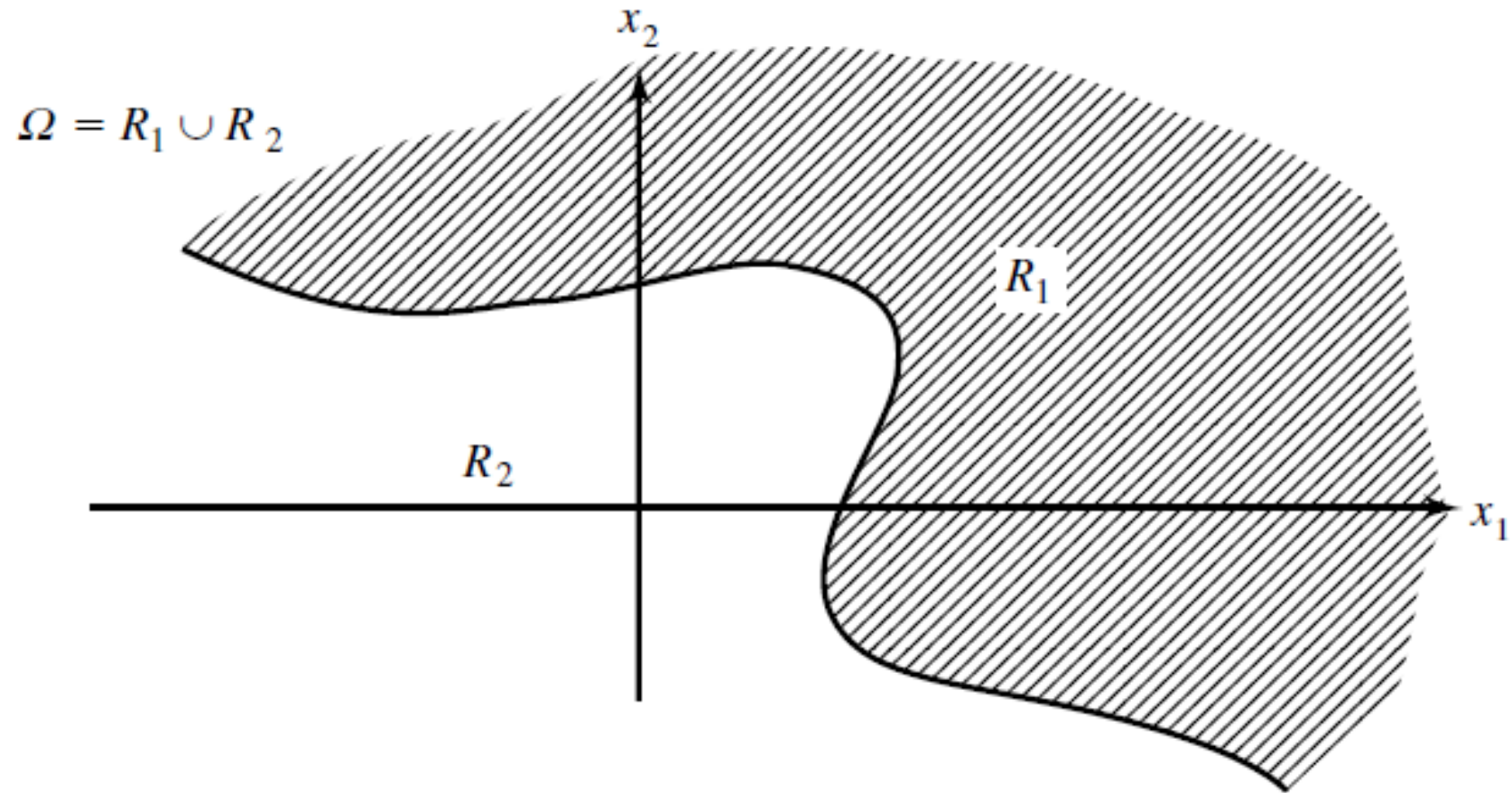
- ❖ These two populations can then be described by probability density functions and
- ❖ Consequently, we can talk of assigning observations to populations or objects to classes interchangeably.

Problems of Misclassification

- ❖ Another aspect of classification is cost.
- ❖ Suppose that classifying an object as belonging to π_2 represents a more serious error than classifying an object as belonging to π_1 , then one should be cautious about making the former assignment.
- ❖ As an example, failing to diagnose a potentially fatal illness is substantially more “costly” than concluding that the disease is present when, in fact, it is not.
- ❖ An optimal classification procedure should, whenever possible, account for the **costs** associated with misclassification.

Probability Distributions

- ❖ Let $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ be the **multivariate probability density functions** associated with the $p \times 1$ vector random variable $\mathbf{X}$ for the populations π_1 and π_2 respectively.
- ❖ An object with associated measurements $\mathbf{x}$ must be assigned to either π_1 or π_2
- ❖ Let Ω be the sample space—that is, the collection of all possible observations $\mathbf{x}$.
- ❖ Let $\mathbf{R}_1$ be that set of $\mathbf{x}$ values for which we classify objects as π_1 and $\mathbf{R}_2 = \Omega - \mathbf{R}_1$ be the set of remaining $\mathbf{x}$ values for which we classify objects as π_2
- ❖ Since every object must be assigned to one and only one of the two populations, the sets $\mathbf{R}_1$ and $\mathbf{R}_2$ are mutually exclusive and exhaustive.



Classification regions for two populations

Probability of Misclassifications

- ❖ The conditional probability, $P(2|1)$ of misclassifying an object as π_2 when, in fact, it is from π_1

$$P(2|1) = P(\mathbf{X} \in R_2 | \pi_1) = \int_{R_2 = \Omega - R_1} f_1(\mathbf{x}) d\mathbf{x}$$

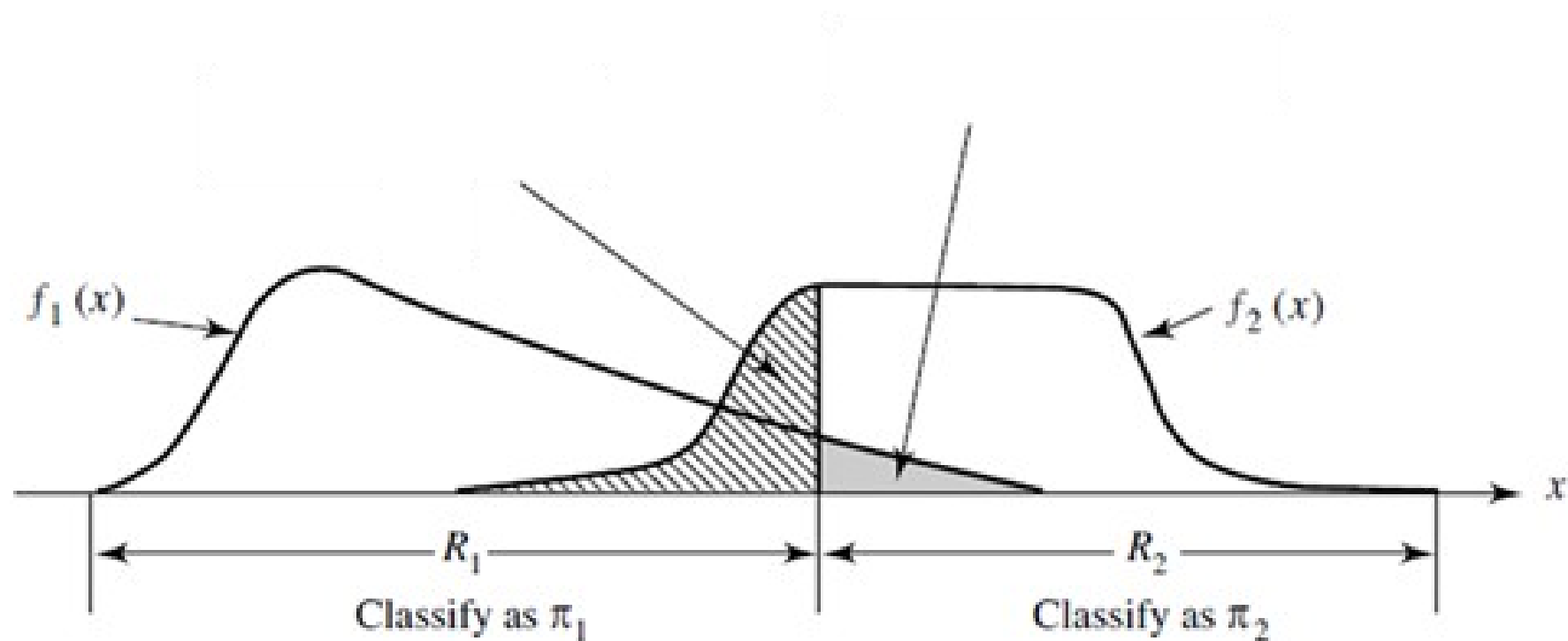
- ❖ Similarly, the conditional probability $P(1|2)$ of misclassifying an object as π_1 when, in fact, it is

$$P(1|2) = P(\mathbf{X} \in R_1 | \pi_2) = \int_{R_1} f_2(\mathbf{x}) d\mathbf{x}$$

from π_2 is

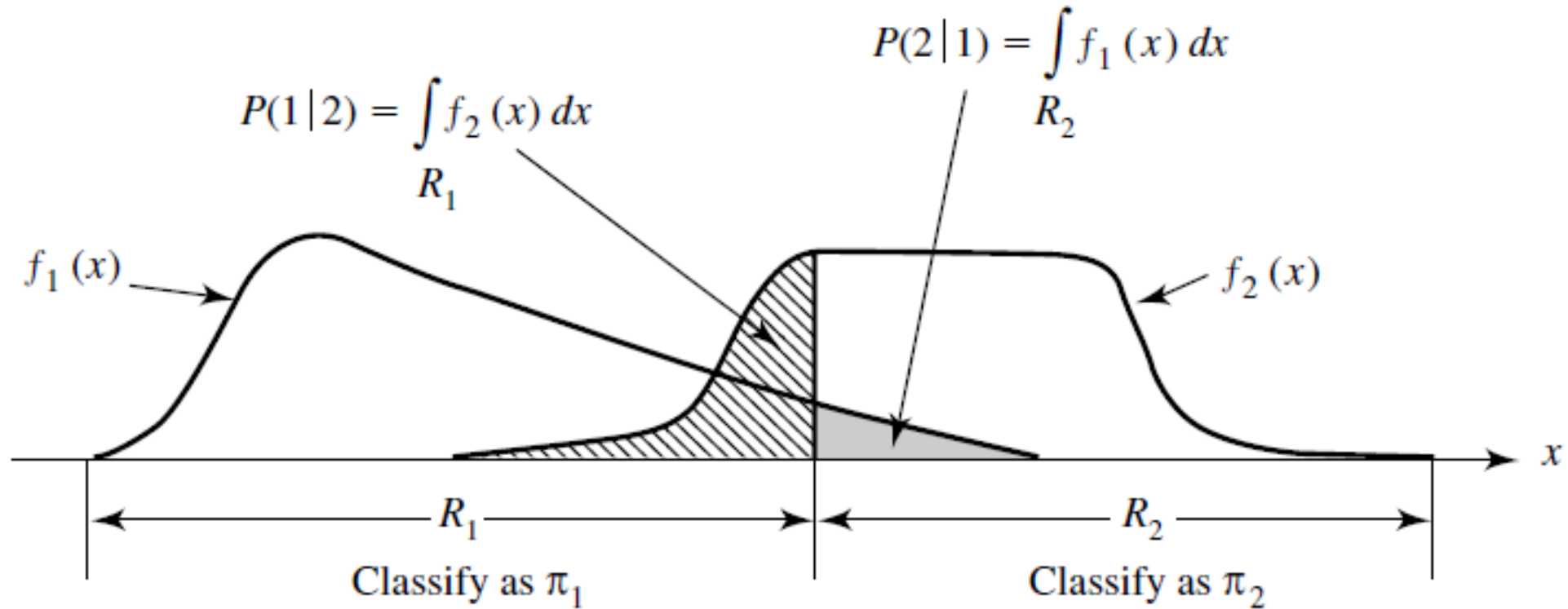
- ❖ The integral sign in first equation represents the volume formed by the density function $f_1(\mathbf{x})$ over the region R_2
- ❖ Similarly, the integral sign in the second equation represents the volume formed by $f_2(\mathbf{x})$ over the

Probability Distributions



Misclassification probabilities for hypothetical classification regions when $p = 1$

Probability Distributions



Misclassification probabilities for hypothetical classification regions when $p = 1$

Probability Distributions

- ❖ Let p_1 be the prior probability of π_1 and p_2 be the prior probability of π_2 , where $p_1 + p_2 = 1$
- ❖ Then the overall probabilities of correctly or incorrectly classifying objects may be derived as the product of the prior and conditional classification probabilities:

$$\begin{aligned} P(\text{observation is correctly classified as } \pi_1) &= P(\text{observation comes from } \pi_1 \\ &\quad \text{and is correctly classified as } \pi_1) \\ &= P(\mathbf{X} \in R_1 | \pi_1)P(\pi_1) = P(1|1)p_1 \end{aligned}$$

Probability Distributions

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$$\begin{aligned}P(\text{observation is misclassified as } \pi_1) &= P(\text{observation comes from } \pi_2 \\ &\quad \text{and is misclassified as } \pi_1) \\ &= P(\mathbf{X} \in R_1 | \pi_2)P(\pi_2) = P(1|2)p_2\end{aligned}$$

$$\begin{aligned}P(\text{observation is correctly classified as } \pi_2) &= P(\text{observation comes from } \pi_2 \\ &\quad \text{and is correctly classified as } \pi_2) \\ &= P(\mathbf{X} \in R_2 | \pi_2)P(\pi_2) = P(2|2)p_2\end{aligned}$$

$$\begin{aligned}P(\text{observation is misclassified as } \pi_2) &= P(\text{observation comes from } \pi_1 \\ &\quad \text{and is misclassified as } \pi_2) \\ &= P(\mathbf{X} \in R_2 | \pi_1)P(\pi_1) = P(2|1)p_1\end{aligned}$$

This is just the multiplication rule for joint probability: $P(A \cap B) = P(A)P(B|A)$.

Why p_2 and not p_1 :

- p_2 = the prior probability that a randomly encountered object actually belongs to population 2 (its true origin).
- p_1 would be the prior probability that an object belongs to population 1 — but that's irrelevant here, because the object in question is *not* from population 1. It just got mistakenly labeled that way.
- p_1 gets used in the *other* error term, where an object truly from population 1 is misclassified into population 2: that term is $p_1 \times P(2|1)$.

So the full Total Probability of Misclassification is:

$$\text{TPM} = p_1 P(2|1) + p_2 P(1|2)$$

Each term pairs the **prior of the true population** with the **conditional error rate for that same true population** — never mixed. You always multiply by the prior of the group the object *actually* belongs to, because that prior tells you how often you'd encounter such an object in the first place, before any classification error even happens.

Cost of Misclassifications

- ❖ Classification schemes are often evaluated in terms of their misclassification probabilities, but this ignores misclassification cost.
- ❖ For example, even a seemingly small probability such as $P(2|1) = 0.06$, may be too large if the cost of making an incorrect assignment to is extremely high.
- ❖ A rule that ignores costs may cause problems
- ❖ The costs of misclassification can be defined by a cost matrix:

		Classify as:	
		π_1	π_2
True population:	π_1	0	$c(2 1)$
	π_2	$c(1 2)$	0

Probability Distributions

- ❖ The costs are (1) zero for correct classification, (2) $c(1|2)$ when an observation from π_2 is incorrectly classified as π_1 (3) $c(2|1)$ when the observation belongs to π_1 but wrongly classified as π_2
- ❖ The average or **expected cost of misclassification (ECM)** is provided by multiplying the off-diagonal entries in the above relations by their probabilities of occurrence. Here,

$$\text{ECM} = c(2|1)P(2|1)p_1 + c(1|2)P(1|2)p_2$$

- ❖ A reasonable classification rule should ensure **ECM is as small as possible**

Explanation Accounting For Costs

❖ The total **COST** of misclassification $T(R, f)$ is given as follows

$$C(1|2)p_2 \int_{R_1} f_2(\mathbf{x})dx + C(2|1)p_1 \int_{R_2} f_1(\mathbf{x})dx = T(R, f)$$

$$C(1|2)p_2 \int_{R_1} f_2(\mathbf{x})dx + C(2|1)p_1(1 - \int_{R_1} f_1(\mathbf{x})dx) = T(R, f)$$

$$C(2|1)p_1 + \left(C(1|2)p_2 \int_{R_1} f_2(\mathbf{x})dx - C(2|1)p_1 \int_{R_1} f_1(\mathbf{x})dx \right) = T(R, f)$$

$$C(2|1)p_1 + \left(\int_{R_1} (C(1|2)p_2 f_2(\mathbf{x}) - C(2|1)p_1 f_1(\mathbf{x})) \right) dx = T(R, f)$$

Explanation Accounting For Costs

$$C(2|1) p_1 + \left(\int_{R_1} (C(1|2)p_2 f_2(\mathbf{x}) - C(2|1)p_1 f_1(\mathbf{x})) \right) dx = T(R, f)$$

❖ The total probability of misclassification $T(R, f)$ is minimized when for all points

in R_1 we get $C(1|2)p_2 f_2(\mathbf{x}) - C(2|1)p_1 f_1(\mathbf{x}) < 0$

Hence, the classification rule becomes : assign $\mathbf{x}$ to π_1 if

$$\frac{f_1(x)}{f_2(x)} > \frac{C(1|2)p_2}{C(2|1)p_1}$$

Explanation Accounting For Costs

$$T(R, f) = C(2|1)p_1 + \int_{R_1} \left[C(1|2)p_2 f_2(x) - C(2|1)p_1 f_1(x) \right] dx$$

This is a functional — its input isn't a number, it's a **set** (R_1 , a subset of the whole sample space). We're minimizing over all possible choices of R_1 . Nothing here says R_1 is fixed in advance; R_1 is literally *the thing we're solving for*. That's the whole point of the exercise: "find the region R_1 that minimizes T ."

And by definition in this whole setup: **** R_1 and R_2 partition the entire space**** — every point x belongs to exactly one of them. So choosing R_1 is the *same act* as choosing, for every single x , whether it's "in R_1 " or "in R_2 ."

Probability Distributions

$$\frac{f_1(x)}{f_2(x)} > \frac{C(1|2)p_2}{C(2|1)p_1}$$

- ❖ When the prior probabilities are unknown, they are often taken to be equal, and the minimum ECM rule involves comparing the ratio of the population densities to the ratio of the appropriate misclassification costs.
- ❖ If the misclassification cost ratio is also indeterminate, it is usually taken to be unity, and the population density ratio is compared with the ratio of the prior probabilities.
- ❖ Note that the prior probabilities are in the reverse order of the densities.
- ❖ Finally, when both the prior probability and misclassification cost ratios are unity, or one ratio is the reciprocal of the other, the optimal classification regions are determined simply by comparing the values of the density functions.

Special Cases of Minimum Expected Cost Regions

(a) $p_2/p_1 = 1$ (equal prior probabilities)

$$R_1: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)}{c(2|1)} \quad R_2: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)}{c(2|1)}$$

(b) $c(1|2)/c(2|1) = 1$ (equal misclassification costs)

$$R_1: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{p_2}{p_1} \quad R_2: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{p_2}{p_1}$$

(c) $p_2/p_1 = c(1|2)/c(2|1) = 1$ or $p_2/p_1 = 1/(c(1|2)/c(2|1))$
(equal prior probabilities and equal misclassification costs)

$$R_1: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq 1 \quad R_2: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < 1$$

Probability Distributions

- ❖ In this case, if $\mathbf{x}_0$ is a new observation and $f_1(\mathbf{x}_0)/f_2(\mathbf{x}_0) \geq 1$, i.e., we assign $\mathbf{x}_0$ to π_1
- ❖ On the other hand if $f_1(\mathbf{x}_0)/f_2(\mathbf{x}_0) < 1$, we assign $\mathbf{x}_0$ to π_2
- ❖ **It is common practice to arbitrarily use case (c) above for classification.**
- ❖ This is tantamount to assuming equal prior probabilities and equal misclassification costs for the minimum ECM rule
- ❖ This is the justification generally provided.
- ❖ It is also equivalent to assuming the prior probability ratio to be the reciprocal of the misclassification cost ratio.

Classification with Two Multivariate Normal Distributions

- ❖ Classification procedures based on normal populations predominate in statistical practice because of their simplicity and reasonably high efficiency across a wide variety of population models.
- ❖ We now assume that $\mathbf{f}_1(x)$ and $\mathbf{f}_2(x)$ are multivariate normal densities, the first with mean vector $\boldsymbol{\mu}_1$ and covariance matrix Σ_1 and the second with mean vector $\boldsymbol{\mu}_2$ and covariance matrix Σ_2
- ❖ The special case of equal covariance matrices leads to a particularly simple linear classification statistic.

= 2
2

Phy	Chem.	Maths
30	45	80
36	51	75
41	36	82

Classification of Normal Populations When $\Sigma_1 = \Sigma_2 = \Sigma$

- ❖ Suppose that the joint densities of $\mathbf{X}$ for populations π_1 and π_2 are given by

$$f_i(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right] \quad \text{for } i = 1, 2$$

- ❖ Let us assume that the population mean vectors $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ as well as Σ are known.

$$R_1: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right)$$

- ❖ Then using the following form

$$\left(\begin{array}{c} \text{density} \\ \text{ratio} \end{array} \right) \geq \left(\begin{array}{c} \text{cost} \\ \text{ratio} \end{array} \right) \left(\begin{array}{c} \text{prior} \\ \text{probability} \\ \text{ratio} \end{array} \right)$$

$$R_2: \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right)$$

$$\left(\begin{array}{c} \text{density} \\ \text{ratio} \end{array} \right) < \left(\begin{array}{c} \text{cost} \\ \text{ratio} \end{array} \right) \left(\begin{array}{c} \text{prior} \\ \text{probability} \\ \text{ratio} \end{array} \right)$$

Classification of Normal Populations When $\Sigma_1 = \Sigma_2 = \Sigma$

$$f_i(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right] \quad \text{for } i = 1, 2$$

❖ Then the expected cost of misclassification becomes

$$\begin{aligned} R_1: \quad & \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_1)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_2)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) \right] \\ & \geq \left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \\ R_2: \quad & \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_1)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_2)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) \right] \\ & < \left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \end{aligned}$$

❖ Given these regions and we can construct the classification rule given in the following result.

Classification of Normal Populations When $\Sigma_1 = \Sigma_2 = \Sigma$

$$\exp \left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)' \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_1) + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_2) \right]$$

❖ Allocate $\mathbf{x}$ to π_1 if

$$(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

❖ Allocate $\mathbf{x}$ to π_2 otherwise

$$R_1: (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

$$R_2: (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) < \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

Classification with Two Multivariate Normal Distributions

- ❖ In most practical situations, the population quantities and are unknown
- ❖ So, the above rule must be modified.
- ❖ It has been suggested replacing the population parameters by their sample counterparts.
- ❖ From the data matrices we compute

$$\begin{aligned} \mathbf{X}_1 &= \begin{bmatrix} \mathbf{x}'_{11} \\ \mathbf{x}'_{12} \\ \vdots \\ \mathbf{x}'_{1n_1} \end{bmatrix} & \bar{\mathbf{x}}_1 &= \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbf{x}_{1j}, & \mathbf{S}_1 &= \frac{1}{n_1 - 1} \sum_{j=1}^{n_1} (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1) (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)' \\ \mathbf{X}_2 &= \begin{bmatrix} \mathbf{x}'_{21} \\ \mathbf{x}'_{22} \\ \vdots \\ \mathbf{x}'_{2n_2} \end{bmatrix} & \bar{\mathbf{x}}_2 &= \frac{1}{n_2} \sum_{j=1}^{n_2} \mathbf{x}_{2j}, & \mathbf{S}_2 &= \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2) (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)' \end{aligned}$$

Classification with Two Multivariate Normal Distributions

- ❖ Since it is assumed that the parent populations have the same covariance matrix Σ the sample covariance matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ are combined (pooled) to derive a single, unbiased estimate of Σ .
- ❖ In particular, the weighted average

$$\mathbf{S}_{\text{pooled}} = \left[\frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_1 + \left[\frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_2$$

is an unbiased estimate of Σ if the data matrices $\mathbf{X}_1$ and $\mathbf{X}_2$ contain random samples from the populations π_1 and π_2 respectively.

Classification with Two Multivariate Normal Distributions

Substituting $\bar{\mathbf{x}}_1$ for $\boldsymbol{\mu}_1$ and $\bar{\mathbf{x}}_2$ for $\boldsymbol{\mu}_2$ and $\mathbf{S}_{\text{pooled}}$ for $\boldsymbol{\Sigma}$ gives the “sample” classification rule:

$$(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} \mathbf{x}_0 - \frac{1}{2} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

The Estimated Minimum ECM Rule for Two Normal Populations

Allocate $\mathbf{x}_0$ to π_1 if

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}_0 - \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

Allocate $\mathbf{x}_0$ to π_2 otherwise.

Classification with Two Multivariate Normal Distributions

The Estimated Minimum ECM Rule for Two Normal Populations

Allocate $\mathbf{x}_0$ to π_1 if

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}_0 - \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

Allocate $\mathbf{x}_0$ to π_2 otherwise.

If we set $\left(\frac{c(1|2)}{c(2|1)} \right) * \left(\frac{p_2}{p_1} \right) = 1$ we find $\ln(1) = 0$ and then the above equation may

be written as two parts after equating the following relation to 0.

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}_0 - \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) = 0$$

Classification with Two Multivariate Normal Distributions

The estimated ECM rule amounts to comparing the scalar variable

$$\hat{y} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x} = \hat{\mathbf{a}}' \mathbf{x}$$

with the
number (second part)

$$\begin{aligned} \hat{m} &= \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \\ &= \frac{1}{2} (\bar{y}_1 + \bar{y}_2) \end{aligned}$$

$$\bar{y}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_1 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_1$$

$$\bar{y}_2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_2 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_2$$

Classification with Two Multivariate Normal Distributions

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}_0 - \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) = 0$$

The estimated ECM rule amounts to comparing the scalar variable

$$\hat{y} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x} = \hat{\mathbf{a}}' \mathbf{x}$$

with the
number (second part)

$$\begin{aligned} \hat{m} &= \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \\ &= \frac{1}{2} (\bar{y}_1 + \bar{y}_2) \end{aligned}$$

$$\bar{y}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_1 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_1$$

$$\bar{y}_2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_2 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_2$$

Probability Distributions

- ❖ That is, the estimated minimum ECM rule for two normal populations is tantamount to creating two univariate populations for the y values.
- ❖ This is done by taking an appropriate linear combination of the observations from populations π_1 and π_2 and then assigning a new observation $\mathbf{x}_o$ to π_1 or π_2 depending upon whether $\hat{y}_o = \hat{\mathbf{a}}'\mathbf{x}_o$ falls to the right or left of the midpoint $\hat{m}$ between the two-univariate means $\bar{y}_1$ and $\bar{y}_2$

Probability Distributions

❖ Once parameter estimates are inserted for the corresponding unknown population quantities, there is no assurance that the resulting rule will minimize the expected cost of misclassification in a particular application

❖ This is because the optimal rule was derived assuming that the multivariate normal densities $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ were known completely

❖ Expression

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}_0 - \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

is simply an estimate of the optimal rule.

❖ It is reasonable to expect that it should perform well if the sample sizes are large

Summary (End of Part 1)

- ❖ To summarize, if the data appear to be multivariate normal, the classification statistic to the left of the inequality can be calculated for each new observation
- ❖ These observations are classified by comparing the values of the statistic with the value of

$$\ln \left[\frac{c(1|2)}{c(2|1)} \left(\frac{p_2}{p_1} \right) \right]$$

Example Problem

- ❖ Using the data given in the excel file do the following
 - a) Carry out the discriminant analysis
 - b) Obtain the sample linear discriminant function, assuming equal prior probabilities
 - c) Classify the following 10 new cases using the discriminant function in Part b.

New Data Set for Classification

New Cases Requiring Classification

Case	$\log_{10}(\text{AHF activity})$	$\log_{10}(\text{AHF antigen})$
1	−.112	−.279
2	−.059	−.068
3	.064	.012
4	−.043	−.052
5	−.050	−.098
6	−.094	−.113
7	−.123	−.143
8	−.011	−.037
9	−.210	−.090
10	−.126	−.019

Summary of Formulae Used

$$\begin{matrix} \bar{\mathbf{x}}_1 \\ (p \times 1) \end{matrix} = \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbf{x}_{1j}, \quad \begin{matrix} \mathbf{S}_1 \\ (p \times p) \end{matrix} = \frac{1}{n_1 - 1} \sum_{j=1}^{n_1} (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)(\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)'$$

$$\begin{matrix} \bar{\mathbf{x}}_2 \\ (p \times 1) \end{matrix} = \frac{1}{n_2} \sum_{j=1}^{n_2} \mathbf{x}_{2j}, \quad \begin{matrix} \mathbf{S}_2 \\ (p \times p) \end{matrix} = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)(\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)'$$

$$\mathbf{S}_{\text{pooled}} = \left[\frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_1 + \left[\frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_2$$

Summary of Formulae Used

$$\hat{y} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x} = \hat{\mathbf{a}}' \mathbf{x}$$

$$\begin{aligned} \hat{m} &= \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \\ &= \frac{1}{2} (\bar{y}_1 + \bar{y}_2) \end{aligned}$$

$$\bar{y}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_1 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_1$$

$$\bar{y}_2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_2 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_2$$

% Problem 11.8 in

clear all

clc

Group1 = [

-0.0056 -0.1657

-0.1698 -0.1585

-0.3469 -0.1879

...

0.1507 0.0933

-0.1259 -0.0669

-0.1551 -0.1232

-0.1952 -0.1007

...

-0.1972 -0.0607

-0.0867 -0.056];

Group2 = [

-0.3478 0.1151

-0.3618 -0.2008

-0.4986 -0.086

...

-0.2444 0.1614

-0.4784 0.0282

];

Matlab Code For Discrimination

div1=length(Group1);

div2=length(Group2);

x1bar=mean(Group1)';

x2bar=mean(Group2)';

S1=cov(Group1);

S2=cov(Group2);

n1m1=div1-1;

n2m1=div2-1;

Spoiled=(n1m1/
(n1m1+n2m1))*S1+

(n2m1/(n1m1+n2m1))*S2;

Spoiledinv=inv(Spoiled);

diffp=(x1bar-x2bar)';

ahatp = diffp*Spoiledinv;

y1bar=ahatp*x1bar;

y2bar=ahatp*x2bar;

mhat=0.5*(y1bar+y2bar);

$$\bar{\mathbf{x}}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbf{x}_{1j}, \quad \mathbf{S}_1 = \frac{1}{n_1 - 1} \sum_{j=1}^{n_1} (\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)(\mathbf{x}_{1j} - \bar{\mathbf{x}}_1)'$$

$$\bar{\mathbf{x}}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} \mathbf{x}_{2j}, \quad \mathbf{S}_2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)(\mathbf{x}_{2j} - \bar{\mathbf{x}}_2)'$$

$$\mathbf{S}_{\text{pooled}} = \left[\frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_1 + \left[\frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_2$$

Matlab Code For Discrimination

```
% Problem 11.8 in
clear all
clc
Group1 = [
-0.0056 -0.1657
-0.1698 -0.1585
-0.3469 -0.1879
...
0.1507 0.0933
-0.1259 -0.0669
-0.1551 -0.1232
-0.1952 -0.1007
...
-0.1972 -0.0607
-0.0867 -0.056 ];
Group2 = [
-0.3478 0.1151
-0.3618 -0.2008
-0.4986 -0.086
...
-0.2444 0.1614
-0.4784 0.0282
];
```

```
div1=length(Group1);
div2=length(Group2);
x1bar=mean(Group1)';
x2bar=mean(Group2)';
S1=cov(Group1);
S2=cov(Group2);
n1m1=div1-1;
n2m1=div2-1;
Spooled=(n1m1/(
n1m1+n2m1))*S1+
(n2m1/(n1m1+n2m1))*S2;
Spooledinv=inv(Spooled);
diffp=(x1bar-x2bar)';
ahatp = diffp*Spooledinv;
y1bar=ahatp*x1bar;
y2bar=ahatp*x2bar;
mhat=0.5*(y1bar+y2bar);
```

$$\mathbf{S}_{\text{pooled}} = \left[\frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_1 + \left[\frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} \right] \mathbf{S}_2$$

$$\hat{y} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x} = \hat{\mathbf{a}}' \mathbf{x}$$

$$\begin{aligned} \hat{m} &= \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \\ &= \frac{1}{2} (\bar{y}_1 + \bar{y}_2) \end{aligned}$$

$$\bar{y}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_1 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_1$$

$$\bar{y}_2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_2 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_2$$

New Data Set for Classification

New Cases Requiring Classification

Case	$\log_{10}(\text{AHF activity})$	$\log_{10}(\text{AHF antigen})$
1	−.112	−.279
2	−.059	−.068
3	.064	.012
4	−.043	−.052
5	−.050	−.098
6	−.094	−.113
7	−.123	−.143
8	−.011	−.037
9	−.210	−.090
10	−.126	−.019

Code For Classification

% Add following lines to previous code

% For classification New Test Data set

Test_Data = [

-0.112 -0.279

-0.059 -0.068

0.064 0.012

-0.043 -0.052

-0.05 -0.098

-0.094 -0.113

-0.123 -0.143

-0.011 -0.037

-0.21 -0.09

-0.126 -0.019];

Result = ahatp*Test_Data';

Score = Result-mhat;

for j=1:length(Test_Data)

if Score(j) > 0

Class(j) = 1;

else

Class(j) = 2;

end

end

Final_Outcome=[Test_Data Score' Class']

Final_Outcome =

-0.1120	-0.2790	6.1734	1.0000
-0.0590	-0.0680	3.5841	1.0000
0.0640	0.0120	4.5904	1.0000
-0.0430	-0.0520	3.6192	1.0000
-0.0500	-0.0980	4.2717	1.0000
-0.0940	-0.1130	3.6785	1.0000
-0.1230	-0.1430	3.6320	1.0000
-0.0110	-0.0370	3.9806	1.0000
-0.2100	-0.0900	1.0437	1.0000
-0.1260	-0.0190	1.4506	1.0000

$$\hat{y} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x} = \hat{\mathbf{a}}' \mathbf{x}$$

$$\begin{aligned} \hat{m} &= \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \\ &= \frac{1}{2} (\bar{y}_1 + \bar{y}_2) \end{aligned}$$

$$\bar{y}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_1 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_1$$

$$\bar{y}_2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{\text{pooled}}^{-1} \bar{\mathbf{x}}_2 = \hat{\mathbf{a}}' \bar{\mathbf{x}}_2$$